another of its constituent parts: its supposed independence from the rest of the world.

The three-handed dance of these intellectual giants—Aristotle, Newton, and Einstein—has guided us to a deeper understanding of time and of space. There is a structure of reality that is the gravitational field; it is not separate from the rest of physics, nor is it the stage across which the world passes. It is a dynamic component of the great dance of the world, similar to all the others, interacting with the others, determining the rhythm of those things that we call meters and clocks and the rhythm of all physical phenomena.

Success, as ever, is destined to be short-lived—even great success. Einstein writes the equations of the gravitational field in 1915, and barely a year later it is Einstein himself who observes that this cannot be the last word on the nature of time and space, because of the existence of quantum mechanics. The gravitational field, like all physical things, must necessarily have quantum properties.